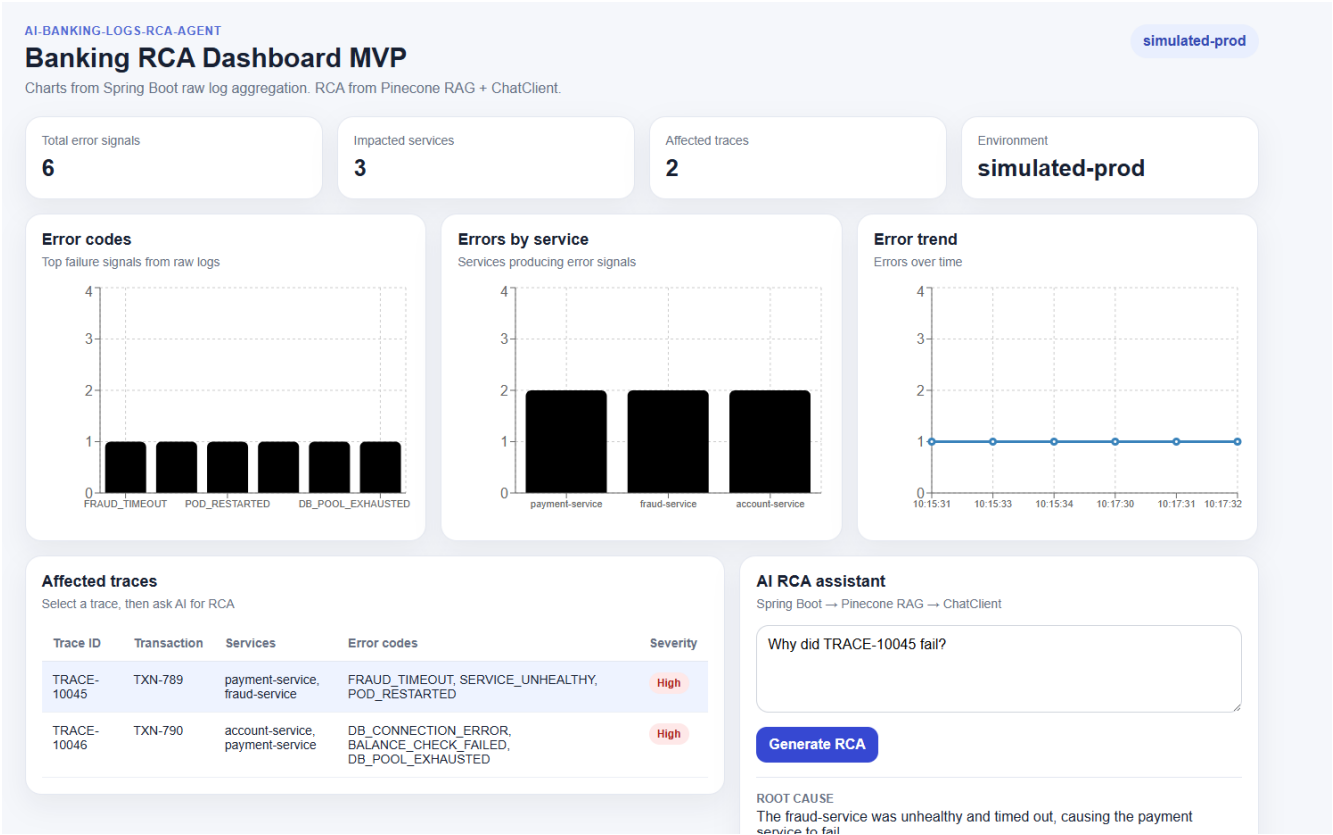


Banking Microservices RCA

AI ProductionSupport Agent



Affected traces

Select a trace, then ask AI for RCA

Trace ID	Transaction	Services	Error codes	Severity
TRACE-10045	TXN-789	payment-service, fraud-service	FRAUD_TIMEOUT, SERVICE_UNHEALTHY, POD_RESTARTED	High
TRACE-10046	TXN-790	account-service, payment-service	DB_CONNECTION_ERROR, BALANCE_CHECK_FAILED, DB_POOL_EXHAUSTED	High

AI RCA assistant

Spring Boot → Pinecone RAG → ChatClient

Why did TRACE-10045 fail?

Generate RCA

ROOT CAUSE

The fraud-service was unhealthy and timed out, causing the payment service to fail.

IMPACTED SERVICE

payment-service

SYMPTOM

Payment failed due to fraud-service timeout.

EVIDENCE USED BY AI

- 2026-06-22T10:15:31Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 | message=Payment failed due to fraud-service timeout
- 2026-06-22T10:15:33Z | service=fraud-service | level=ERROR | eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY | latencyMs=4200 | message=Readiness probe failed: /health returned 503
- 2026-06-22T10:15:34Z | service=fraud-service | level=WARN | eventType=POD_RESTARTED | errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated health check failures

SUGGESTED FIX

- Investigate the health of the fraud-service and resolve any underlying issues.
- Monitor the readiness probe configuration and adjust it if necessary.

► Retrieved Pinecone evidence

Steps to run

Start watch_raw_logs_to_cloudwatch.py

```
inal Help < > AIEngineering_JavaPython 1
```

```
generate_logs.py U watch_raw_logs_to_cloudwatch.py U X
```

```
AI-Banking-Logs-RCA-Agent > RAG-Ingestion > automation > watch_raw_logs_to_cloudwatch.py > ...
```

```
1 import subprocess
2 import sys
3 import time
4 from pathlib import Path
5
6 from watchdog.events import FileSystemEventHandler
7 from watchdog.observers import Observer
8
9
10 BASE_DIR = Path(__file__).resolve().parent.parent
11 RAW_LOGS_FILE = BASE_DIR / "data" / "raw_logs.jsonl"
12 PUSH_SCRIPT = BASE_DIR / "cloud" / "push_logs_to_cloudwatch.py"
13
14
15 class RawLogsChangeHandler(FileSystemEventHandler):
16     def __init__(self):
17         self.last_run_time = 0
18
19     def on_modified(self, event):
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
```

```
ope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\saito\source\repos\AIEngineering_JavaPython\Langchain\RAG_LC\.venv\Scripts\Activate.ps1)
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython> & c:\Users\saito\source\repos\AIEngineering_JavaPython\Langchain\RAG_LC\.venv\Scripts\python.exe c:/Users/saito/source/repos/AIEngineering_JavaPython/AI-Banking-Logs-RCA-Agent/RAG-Ingestion/log-generator/generate_logs.py
Generated 15 logs
Saved to C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent\RAG-Ingestion\data\raw_logs.jsonl
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython> & c:\Users\saito\source\repos\AIEngineering_JavaPython\Langchain\RAG_LC\.venv\Scripts\python.exe c:/Users/saito/source/repos/AIEngineering_JavaPython/AI-Banking-Logs-RCA-Agent/RAG-Ingestion/automation/watch_raw_logs_to_cloudwatch.py
Watching raw logs file:
C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent\RAG-Ingestion\data\raw_logs.jsonl
Press Ctrl+C to stop.
```

```
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython> & c:\Users\saito\source\repos\AIEngineering_JavaPython\Langchain\RAG_LC\.venv\Scripts\python.exe c:/Users/saito/source/repos/AIEngineering_JavaPython/AI-Banking-Logs-RCA-Agent/RAG-Ingestion/automation/watch_raw_logs_to_cloudwatch.py
Watching raw logs file:
C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent\RAG-Ingestion\data\raw_logs.jsonl
Press Ctrl+C to stop.
```

Start Frontend

```
npm run dev
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent\
Frontend\rca-dashboard-final> npm run dev

> dev
> vite

VITE v8.1.0 ready in 934 ms

  → Local:   http://localhost:5173/
  → Network: use --host to expose
  → press h + enter to show help
3:29:44 PM [vite] (client) [optimizer] bundling dependencies...
c:\Users\saito\source\repos\AIEngineering_JavaPython\Langchain\RAG_LC\.venv\Scripts\activa
te
□
```

npm run dev

Run generate_logs.py

Start Java Backend

The screenshot shows an IDE with the following code in `RcaBackendApplication.java`:

```
1 package com.banking.rca_backend;
2
3 import org.springframework.boot.SpringApplication;
4
5
6 @SpringBootApplication
7 public class RcaBackendApplication {
8
9     public static void main(String[] args) {
10         SpringApplication.run(RcaBackendApplication.class, args);
11     }
12
13 }
14
```

The console output shows the application starting successfully:

```
:: Spring Boot :: (v3.5.15)

2026-06-24T15:40:45.991-04:00 INFO 100180 --- [rca-backend] [main] c.b.rca_backend.RcaBackendApplication : Starting
2026-06-24T15:40:46.000-04:00 INFO 100180 --- [rca-backend] [main] c.b.rca_backend.RcaBackendApplication : No active
2026-06-24T15:40:50.195-04:00 INFO 100180 --- [rca-backend] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat in
2026-06-24T15:40:50.227-04:00 INFO 100180 --- [rca-backend] [main] o.apache.catalina.core.StandardService : Starting
2026-06-24T15:40:50.228-04:00 INFO 100180 --- [rca-backend] [main] o.apache.catalina.core.StandardEngine : Starting
2026-06-24T15:40:50.336-04:00 INFO 100180 --- [rca-backend] [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializ
2026-06-24T15:40:50.337-04:00 INFO 100180 --- [rca-backend] [main] w.s.c.ServletWebServerApplicationContext : Root WebA
2026-06-24T15:40:53.243-04:00 INFO 100180 --- [rca-backend] [main] o.s.b.a.e.web.EndpointLinksResolver : Exposing
2026-06-24T15:40:53.387-04:00 INFO 100180 --- [rca-backend] [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat st
2026-06-24T15:40:53.420-04:00 INFO 100180 --- [rca-backend] [main] c.b.rca_backend.RcaBackendApplication : Started R
2026-06-24T15:41:05.732-04:00 INFO 100180 --- [rca-backend] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializ
2026-06-24T15:41:05.733-04:00 INFO 100180 --- [rca-backend] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializ
2026-06-24T15:41:05.738-04:00 INFO 100180 --- [rca-backend] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed
```

I run Python generate_logs

```
\RAG-Ingestion> cd .\log-generator\
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent
\RAG-Ingestion\log-generator> python .\generate_logs.py
Generated 15 logs
Saved to C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent\RA
G-Ingestion\data\raw_logs.jsonl
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent
\RAG-Ingestion\log-generator>
```

(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython\AI-Banking-Logs-RCA-Agent\RAG-Ingestion\log-generator> python .\generate_logs.py
Generated 15 logs

raw_logs.jsonl is updated

```
Help  ← →  AIEngineering_JavaPython  1  ↕  ↗

generate_logs.py U  {} raw_logs.jsonl U X  watch_raw_logs_to_cloudwatch.py U  ↻  🔍

AI-Banking-Logs-RCA-Agent > RAG-Ingestion > data > {} raw_logs.jsonl
1  {"timestamp": "2026-06-22T10:15:29Z", "environment": "simulated-prod", "traceId": "TRACE-10045"}
2  {"timestamp": "2026-06-22T10:15:30Z", "environment": "simulated-prod", "traceId": "TRACE-10045"}
3  {"timestamp": "2026-06-22T10:15:31Z", "environment": "simulated-prod", "traceId": "TRACE-10045"}
4  {"timestamp": "2026-06-22T10:15:33Z", "environment": "simulated-prod", "traceId": "TRACE-10045"}
5  {"timestamp": "2026-06-22T10:15:34Z", "environment": "simulated-prod", "traceId": "TRACE-10045"}
6  {"timestamp": "2026-06-22T10:17:29Z", "environment": "simulated-prod", "traceId": "TRACE-10046"}
7  {"timestamp": "2026-06-22T10:17:30Z", "environment": "simulated-prod", "traceId": "TRACE-10046"}
8  {"timestamp": "2026-06-22T10:17:31Z", "environment": "simulated-prod", "traceId": "TRACE-10046"}
9  {"timestamp": "2026-06-22T10:17:32Z", "environment": "simulated-prod", "traceId": "TRACE-10046"}
10 {"timestamp": "2026-06-22T10:20:10Z", "environment": "simulated-prod", "traceId": "TRACE-10047"}
11 {"timestamp": "2026-06-22T10:20:11Z", "environment": "simulated-prod", "traceId": "TRACE-10047"}
12 {"timestamp": "2026-06-22T10:23:40Z", "environment": "simulated-prod", "traceId": "TRACE-10048"}
13 {"timestamp": "2026-06-22T10:23:41Z", "environment": "simulated-prod", "traceId": "TRACE-10048"}
14 {"timestamp": "2026-06-22T10:25:10Z", "environment": "simulated-prod", "traceId": "TRACE-10049"}
15 {"timestamp": "2026-06-22T10:25:11Z", "environment": "simulated-prod", "traceId": "TRACE-10049"}
16
```

I go to Pinecone

https://app.pinecone.io/organizations/-OvrIsA_yJ8jfTYVBkou/projects/c3748f4f-3842-4b5d-8e41-ec634f41c7ae/indexes/banking-logs-rca/browser

I see everything in the “simulated-prod” Namespace

Namespace	Operation	Text
simulated-prod	Search by text ▼	Hello, world!

+ Filter + Rerank

Search: 5 results (top_k=10)

1	_id: trace-TRACE-10045 environment: "simulated-prod" errorCodes: "FRAUD_TIMEOUT,POD_RESTARTED,SERVICE_UNHEALTHY" services: "account-service,auth-service,fraud-service,payment-service" text: "Evidence logs for traceld TRACE-10045 and transactionId TXN-789.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z service=auth-se traceld: "TRACE-10045" transactionId: "TXN-789"
SCORE 0.0587	
2	_id: trace-TRACE-10046 environment: "simulated-prod" errorCodes: "BALANCE_CHECK_FAILED,DB_CONNECTION_ERROR,DB_POOL_EXHAUSTED" services: "account-service,auth-service,payment-service" text: "Evidence logs for traceld TRACE-10046 and transactionId TXN-790.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z service=auth-se traceld: "TRACE-10046" transactionId: "TXN-790"
SCORE 0.0567	
3	_id: trace-TRACE-10048 environment: "simulated-prod" errorCodes: "QUEUE_PUBLISH_FAILED" services: "notification-service,payment-service" text: "Evidence logs for traceld TRACE-10048 and transactionId TXN-792.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:23:40Z service=payme traceld: "TRACE-10048"
SCORE 0.0182	

Next go to AWS CloudWatch => Log management => /banking/simulated-prod => local-log-generator

EC2 Aurora and RDS S3 Billing and Cost Manage... IAM Elastic Beanstalk Lambda CloudWatch Simple Notification Service VPC EFS Cloud

CloudWatch > Log management > /banking/simulated-prod > local-log-generator

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Timestamp	Message
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:17:33.000Z	{ "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WARN", "traceId": "TRACE-10046", "transactionId": "TXN-791", "environment": "simulated-prod" }
2026-06-22T10:20:10.000Z	{ "timestamp": "2026-06-22T10:20:10Z", "environment": "simulated-prod", "traceId": "TRACE-10047", "transactionId": "TXN-791", "eventType": "APPLICATION_LOG", "message": "Token validation successful", "errorCode": "NONE", "latencyMs": 120, "environment": "simulated-prod" }
2026-06-22T10:20:10.000Z	{ "timestamp": "2026-06-22T10:20:10Z", "environment": "simulated-prod", "traceId": "TRACE-10047", "transactionId": "TXN-791", "eventType": "APPLICATION_LOG", "message": "Token validation successful", "errorCode": "NONE", "latencyMs": 120, "environment": "simulated-prod" }
2026-06-22T10:20:11.000Z	{ "timestamp": "2026-06-22T10:20:11Z", "environment": "simulated-prod", "traceId": "TRACE-10047", "transactionId": "TXN-791", "eventType": "APPLICATION_LOG", "message": "Token validation successful", "errorCode": "NONE", "latencyMs": 120, "environment": "simulated-prod" }
2026-06-22T10:20:11.000Z	{ "timestamp": "2026-06-22T10:20:11Z", "environment": "simulated-prod", "traceId": "TRACE-10047", "transactionId": "TXN-791", "eventType": "APPLICATION_LOG", "message": "Token validation successful", "errorCode": "NONE", "latencyMs": 120, "environment": "simulated-prod" }

There are older events to load. [Load more.](#)

2026-06-22T10:15:29.000Z { "timestamp": "2026-06-22T10:15:29Z", "service": "auth-service", "level": "INFO", "traceId": "TRACE-10045", "transactionId": "TXN-789", "eventType": "APPLICATION_LOG", "message": "Token validation successful", "errorCode": "NONE", "latencyMs": 120, "environment": "simulated-prod" }

Raw logs are uploaded into CloudWatch then Lambda applies Chunking and groups raw_logs into groups by TraceID.

CloudWatch raw logs

- Lambda receives batch
- parses each JSON log line
- groups logs by traceId
- creates 1 evidence document per traceId
- upserts that document to Pinecone

You can go to Log management => aws/lambda/banking-ingestion-lambda and pick the latest log

Log events

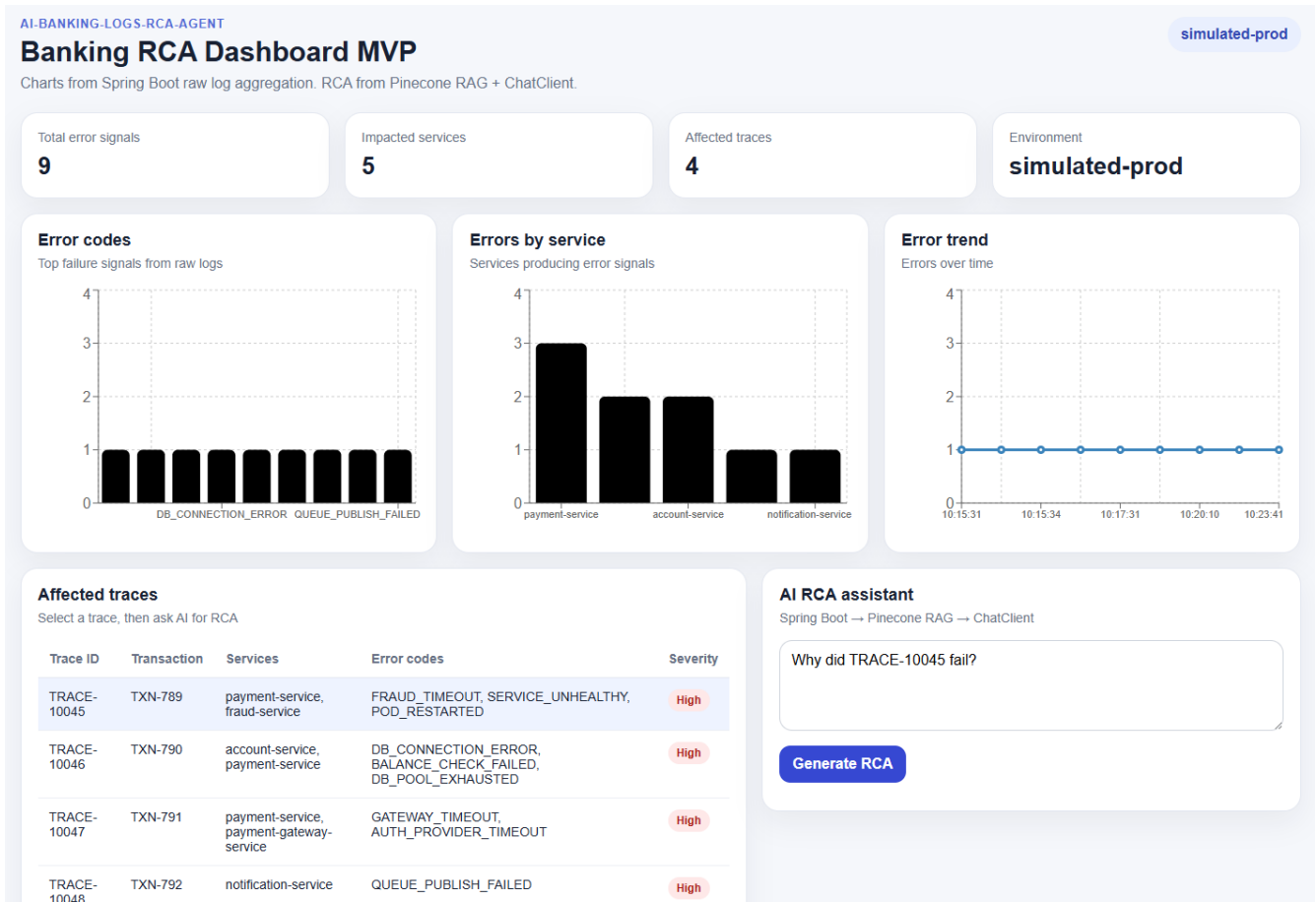
You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q Filter events - press enter to search

▶	Timestamp	Message
▶	2026-06-24T19:47:15.216Z	Pinecone response body:
▶	2026-06-24T19:47:15.216Z	END RequestId: c3c08fe2-bac7-442e-bb06-a69567ecd801
▶	2026-06-24T19:47:15.216Z	REPORT RequestId: c3c08fe2-bac7-442e-bb06-a69567ecd801 Duration: 308.02 ms Billed Duration: 308.02 ms Memory Size: 128 MB Max Memory Used: 128 MB
▶	2026-06-24T19:50:48.162Z	START RequestId: 0b76a522-a3e8-492b-a4d9-08092ab34495 Version: \$LATEST
▶	2026-06-24T19:50:48.174Z	REST PINECONE INGESTION VERSION IS RUNNING
▶	2026-06-24T19:50:48.214Z	Decoded CloudWatch payload successfully
▶	2026-06-24T19:50:48.214Z	Parsed 15 JSON log messages
▶	2026-06-24T19:50:48.214Z	Created 5 evidence records
▶	2026-06-24T19:50:48.815Z	Pinecone response status: 201
▶	2026-06-24T19:50:48.815Z	Pinecone response body:
▶	2026-06-24T19:50:48.817Z	END RequestId: 0b76a522-a3e8-492b-a4d9-08092ab34495
▶	2026-06-24T19:50:48.817Z	REPORT RequestId: 0b76a522-a3e8-492b-a4d9-08092ab34495 Duration: 653.69 ms Billed Duration: 653.69 ms Memory Size: 128 MB Max Memory Used: 128 MB
▶	2026-06-24T19:50:49.947Z	START RequestId: aebf418e-9ce0-44cb-a049-98a354dff4cf Version: \$LATEST
▶	2026-06-24T19:50:49.948Z	REST PINECONE INGESTION VERSION IS RUNNING
▶	2026-06-24T19:50:49.948Z	Decoded CloudWatch payload successfully
▶	2026-06-24T19:50:49.948Z	Parsed 15 JSON log messages
▶	2026-06-24T19:50:49.948Z	Created 5 evidence records
▶	2026-06-24T19:50:50.520Z	Pinecone response status: 201
▶	2026-06-24T19:50:50.520Z	Pinecone response body:

Lambda chunks all raw data by TraceID into Evidence record by TraceIDs

Now we go to Frontend: <http://localhost:5173/>



Prompt

The dashboard shows repeated payment-service errors. Is payment-service the real root cause, or are upstream dependencies causing the failures?

Root cause

Upstream dependencies (specifically payment-gateway-service and fraud-service) are causing the failures.

Impacted service

payment-service

Symptom

Payment authorization and payment processing errors in the payment-service.

Evidence used by AI

Payment authorization failed because payment-gateway-service timed out (GATEWAY_TIMEOUT)

Payment failed due to fraud-service timeout (FRAUD_TIMEOUT)

Suggested fix

Investigate and resolve timeout issues with payment-gateway-service.

Address the health and readiness of the fraud-service to prevent service unavailability.

Retrieved Pinecone evidence

```
[
  "ID: trace-TRACE-10047\nScore: 0.3562491834163666\nTrace ID: TRACE-10047\nTransaction ID:
TXN-791\nServices: payment-gateway-service,payment-service\nError Codes:
AUTH_PROVIDER_TIMEOUT,GATEWAY_TIMEOUT\nEnvironment: simulated-prod\n\nEvidence:
\nEvidence logs for traceId TRACE-10047 and transactionId TXN-791.\nThe following logs show the
request flow in timestamp order:\n\n2026-06-22T10:20:10Z | service=payment-service | level=ERROR
| eventType=APPLICATION_ERROR | errorCode=GATEWAY_TIMEOUT | latencyMs=6200 |
message=Payment authorization failed because payment-gateway-service timed out\n2026-06-
22T10:20:11Z | service=payment-gateway-service | level=ERROR |
eventType=DOWNSTREAM_TIMEOUT | errorCode=AUTH_PROVIDER_TIMEOUT |
latencyMs=6100 | message=External authorization provider did not respond within timeout\n",
  "ID: trace-TRACE-10045\nScore: 0.32967695593833923\nTrace ID: TRACE-10045\nTransaction
ID: TXN-789\nServices: account-service,auth-service,fraud-service,payment-service\nError Codes:
FRAUD_TIMEOUT,POD_RESTARTED,SERVICE_UNHEALTHY\nEnvironment: simulated-
prod\n\nEvidence:\nEvidence logs for traceId TRACE-10045 and transactionId TXN-789.\nThe
following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z | service=auth-
service | level=INFO | eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=120 |
message=Token validation successful\n2026-06-22T10:15:30Z | service=account-service | level=INFO
| eventType=APPLICATION_LOG | errorCode=NONE | latencyMs=210 | message=Account balance
check successful\n2026-06-22T10:15:31Z | service=payment-service | level=ERROR |
eventType=APPLICATION_ERROR | errorCode=FRAUD_TIMEOUT | latencyMs=4300 |
message=Payment failed due to fraud-service timeout\n2026-06-22T10:15:33Z | service=fraud-service
| level=ERROR | eventType=READINESS_PROBE_FAILED | errorCode=SERVICE_UNHEALTHY |
latencyMs=4200 | message=Readiness probe failed: /health returned 503\n2026-06-22T10:15:34Z |
service=fraud-service | level=WARN | eventType=POD_RESTARTED |
errorCode=POD_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated
health check failures\n"
]
```

10:15:31

AI RCA assistant

Spring Boot → Pinecone RAG → ChatClient

The dashboard shows repeated payment-service errors. Is payment-service the real root cause, or are upstream dependencies causing the failures?

Generate RCA

ROOT CAUSE

Upstream dependencies (specifically payment-gateway-service and fraud-service) are causing the failures.

IMPACTED SERVICE

payment-service

SYMPTOM

Payment authorization and payment processing errors in the payment-service.

EVIDENCE USED BY AI

- Payment authorization failed because payment-gateway-service timed out (GATEWAY_TIMEOUT)
- Payment failed due to fraud-service timeout (FRAUD_TIMEOUT)

SUGGESTED FIX

- Investigate and resolve timeout issues with payment-gateway-service.
- Address the health and readiness of the fraud-service to prevent service unavailability.

▼ Retrieved Pinecone evidence

```
[
  "ID: trace-TRACE-10047\nScore: 0.3562491834163666\nTrace ID: TRACE-10047\nTransaction ID: TXN-791\nServices: payment-gateway-service,payment-service\nError Codes: AUTH_PROVIDER_TIMEOUT,GATEWAY_TIMEOUT\nEnvironment: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10047 and transactionId TXN-791.\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:20:10Z | service=payment-service | level=ERROR | eventType=APPLICATION_ERROR | errorCode=GATEWAY_TIMEOUT | latencyMs=6200 | message=Payment authorization failed because payment-gateway-service timed out\n2026-06-22T10:20:11Z | service=payment-gateway-service | level=ERROR | eventType=DOWNSTREAM_TIMEOUT | errorCode=AUTH_PROVIDER_TIMEOUT | latencyMs=6100 | message=External authorization provider did not respond within timeout\n",
  "ID: trace-TRACE-10045\nScore: 0.32967695593833923\nTrace ID: TRACE-10045\nTransaction ID: TXN-789\nServices: account-service,auth-service,fraud-service,payment-service\nError Codes: FRAUD_TIMEOUT,POD_RESTARTED,SERVICE_UNHEALTHY\nEnvironment: simulated-prod\n\nEvidence:\nEvidence logs for traceId TRACE-10045 and transactionId TXN-789.\nThe following logs show the request flow in timestamp
```

Prompt
why trace-TRACE-10047 fail?

Root cause
Database connection issues leading to account balance verification failure.

Impacted service
payment-service

Symptom
Payment failed because account balance could not be verified.

Evidence used by AI
Account balance lookup failed because database connection timed out.
Payment failed because account balance could not be verified.
Connection pool exhausted while connecting to account database.
Suggested fix
Investigate and resolve the database connection timeout issue.
Increase the database connection pool size to prevent exhaustion.

AI RCA assistant

Spring Boot → Pinecone RAG → ChatClient

why trace-TRACE-10047 fail?

Generate RCA

ROOT CAUSE

Database connection issues leading to account balance verification failure.

IMPACTED SERVICE

payment-service

SYMPTOM

Payment failed because account balance could not be verified.

EVIDENCE USED BY AI

- Account balance lookup failed because database connection timed out.
- Payment failed because account balance could not be verified.
- Connection pool exhausted while connecting to account database.

SUGGESTED FIX

- Investigate and resolve the database connection timeout issue.
- Increase the database connection pool size to prevent exhaustion.

.env in Python

```
OPENAI_API_KEY=your_openai_api_key
PINECONE_API_KEY=*****
PINECONE_INDEX_NAME=*****
PINECONE_NAMESPACE=simulated-prod
```

```
AWS_ACCESS_KEY_ID=*****
AWS_SECRET_ACCESS_KEY=*****
AWS_REGION=ca-central-1
```

.env in Java

```
OPENAI_API_KEY=*****
PINECONE_API_KEY=&*****
PINECONE_INDEX_HOST=*****
PINECONE_NAMESPACE=simulated-prod
```

Environment variables in Lambda

Environment variables (4)

The environment variables below are encrypted

 Search Environment variables

Key

PINECONE_API_KEY

PINECONE_INDEX_HOST

PINECONE_INDEX_NAME

PINECONE_NAMESPACE